



The impact of river sand mining and waste management on the Volta Lake: A case study of Asutsuare

Daniel Kwaku Amoah^{a,*}, Esi Awuah^b, Isaac O.A. Hodgson^a,
Miriam Appiah-Brempong^b, Elizabeth Von-Kiti^c, Isaac Nii Yartey Nyarko^d

^a Environmental Chemistry and Sanitation Engineering Division, CSIR-Water Research Institute, Accra, Ghana

^b Regional Water and Environmental Sanitation Center, KNUST, Kumasi, Ghana

^c CSIR- Institute of Industrial Research, Accra, Ghana

^d Osudoku Water Supply Systems, Community Water and Sanitation Agency, Asutsuare, Ghana

ARTICLE INFO

Editor: DR B Gyampoh

Keywords:

Water quality

Sand mining

Volta river

Biochar

Physico-chemical

ABSTRACT

The safety of drinking water is becoming more crucial as the world's population is increasing and anthropogenic activities which pollute water resources are intensifying. This study was carried out between June and August of 2022 to determine the physicochemical and microbial parameters of the Volta Lake at Asutsuare due to sand mining activities and to evaluate how well coconut husk biochar could remove these contaminants. Water samples were taken from one meter below the surface at the upstream, midstream (sand mining site) and downstream once a week for three months. Water samples from the midstream were passed through a coconut husk biochar column to determine its ability to remove pollutants in the water samples. Compared to the upstream, there was a significant difference ($p < 0.05$) in colour, turbidity, iron, manganese, biochemical oxygen demand, chemical oxygen demand, total coliform and total suspended solids from the water sampled from the midstream and downstream sites. Moreover, these values also exceeded the WHO guidelines. There were no significant differences ($p > 0.05$) between the midstream, the upstream and downstream sampling sites for the following parameters: zinc, lead, copper, conductivity, TDS, temperature, faecal coliform, *Escherichia coli* and pH. Biochar from coconut husk was used once and seems to be a highly effective sorbent for removing pollutants from water samples (>70 % removal efficiency).

Introduction

As the world population grows and anthropogenic activities intensify, pollution of water resources is on the rise. According to the World Health Organization (WHO), one in three individuals in rural regions do not utilise clean, safe drinking water, and 2.1 billion people (or 29 % of the global population) lack access to such resources [1]. Additionally, 159 million people continue to obtain their drinking water from surface water sources [1]. "Clean Water and Sanitation" is one of 17 interconnected global goals that are expected to be achieved by 2030, according to the United Nations Sustainable Development Goals (SDGs) [1]. A major concern with surface water is the presence of organic, inorganic, and microbial contaminants, exposure to these substances can result in several health problems. Carcinogenic metals have been found in drinking water [2]. The use of Arsenic-contaminated drinking water by villagers in

* Corresponding author.

E-mail address: kwakudaniel@live.com (D.K. Amoah).

<https://doi.org/10.1016/j.sciaf.2024.e02359>

Received 17 April 2024; Received in revised form 13 August 2024; Accepted 13 September 2024

Available online 18 September 2024

2468-2276/© 2024 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

India has been linked to severe health effects [3].

Overview of sand mining

Historically, sand mining has been practiced in many parts of the world. Many countries worldwide, including Kenya, Ghana, India, and Malaysia, have laws to regulate sand mining and harvesting due to the environmental and socioeconomic effects [4]. Policy implementation issues in other nations including India lead to unauthorized and unrestricted sand mining [5]. Indiscriminate sand mining has also been going on in Ghana, where it's being done in an unsustainable and unregulated way [6]. In Botswana and South Africa the extraction of sand can be done on a small as well as a large scale [7].

Sand mining in Ghana is done in the Coastal Parts of the country [6]. Sand mining that takes place on the terrace, flood plain, or overbank is referred to as off-channel sand mining while In-channel sand sources are associated with in-stream sand mining, which involves taking sand from a river that is currently flowing. Natural sand may be found in other locations, for example, lakes, coastlines, water reservoirs, marine sources, and seashore areas. These are referred to as non-river sources of sand, even though they can additionally be utilized for similar reasons as river sand [8].

Types of sand harvesting

Sand mining can have an impact on rivers in various ways, depending upon the type of mining performed. The activities of riverine aggregate mining can generally be divided into three categories; wet mining, dry mining, and bar skimming or scalping. Typically, in-stream mining consists of mining aggregates directly from the bed of flowing rivers. Such mining can take place either within the channel or near the margins of a river. On floodplains, at the margins of seasonal rivers, or during periods when rivers are dry throughout the year, dry mining is conducted. In bar skimming or scalping, the top level of exposed sand bars is removed (to varying depths) [9]. In-stream sand mining normally occurs internally in a flowing river which is the preferred approach of mining sand since it is less expensive. As a result, the aquatic ecosystem is affected both qualitatively and quantitatively [8]. Mining for river sand may produce large amounts of suspended particles, which can have a negative impact on the environment and water quality of rivers [10].

Effect of sand mining on water quality

A key component of global sustainability improvement is water quality. As a result of human activity near and inside water bodies, where it is worsening every day, water pollutants have turned out to be a problem worldwide [11]. The biological and physicochemical components of water determine how well an aquatic and riparian environment is protected. The declining quality of water additionally influences human fitness and socioeconomic popularity [12]. The specific levels of carbons, vitamins, as well as ions below-average levels seem to be massive and ought to continue to be at stable ranges for aquatic life to be able to survive in water [11]. However, we must not disregard the effects that human activity has on rivers and other bodies of water. Land operations such as sand mining in various regions of the sector have disrupted and altered the biological and physicochemical components of the water flowing in the rivers, which has an impact on the river water quality [13].

Human activities, such as riparian land alterations and the mining of sand and gravel soils, which can cause erosion and soil entering water bodies, may be the source of the turbidity of the water. The acidity or alkalinity of the water may have an impact on the wellbeing of aquatic species and the pH of water affects its suitability [11–13]. The pH of the river may be impacted by the mining of river sand using rusty equipment. The organisms in the river will eventually die as they are unable to adjust to the changes in pH. [11]. Changes in turbidity, temperature and habitat distribution are among the effects of gravel and sand mining on water quality. Water pollution and saltwater intrusion are also a result of river sand mining [14].

River water quality is reduced by pollution due to the introduction of microorganisms that can cause waterborne diseases in humans and alter the physicochemical composition, rendering it unsafe for aquatic life to survive [15].

Besides affecting pH, BOD, COD and turbidity of river water, sand harvesting increases temperature, conductivity, and the presence of heavy metals [11]. Humans and aquatic life are at risk when these water properties are altered [16].

Biochar production for water quality

Biochar can be produced from a wide range of biomass sources, including waste materials, wood, animal dung, and agricultural residues [17]. Hardwood or softwood species are frequently used to make wood-based biochars, which are renowned for their stability and high carbon content. A natural adsorbent that is inexpensive and widely accessible is agricultural waste [18]. Due to its lingo-cellulosic content, numerous studies suggest its effective removal of synthetic antibiotics, heavy metals, and pesticides. Vegetable and fruit peels, sawdust, leaves, husks, shells, bran, and straw are a few common agricultural waste materials utilized as adsorbents for water treatment [19]. Techniques for water filtration using biochar have received a lot of attention recently [20]. Furthermore, using biochar to purify drinking water is essential owing to its cost-effectiveness and environmental friendliness [21]. Coconut husk and coir have beneficial properties such as structural stability, high absorptivity, and porous structure that make them excellent bio-adsorbents for wastewater contaminants. Physical and chemical adsorption are the two primary adsorption mechanisms of biochar for pollutants. Van der Waals force or specific surface area is the primary basis for physical adsorption [20]. The primary factors considered in biochar-based water filtration for microbiological pollutant removal are the type of harmful bacteria, their features, pathogenic levels in the water, the volume of biochar employed, and procedures [22]. Biochar offers several potential

advantages when compared to current low-cost techniques. Low-income communities can benefit from biochar since it is an affordable, renewable adsorbent that is produced with easily accessible biomaterials and skills. While biochar removes biological, physical, and chemical pollutants, existing approaches primarily eliminate infections. While current techniques produce hazardous by-products, biochar preserves the organoleptic qualities of water [23]. Biochar offers many benefits, but it also has a lot of drawbacks. Pure biochar has a relatively low adsorption capacity for anionic contaminants, but an outstanding adsorption capability for organic compounds. Furthermore, because of its porous structure and small number of surface functional groups, raw biochar takes a long time to reach equilibrium [24].

Besides affecting the quality of the river water and groundwater recharge, excessive sand mining damages aquatic habitats and microorganisms. High levels of siltation caused by sand mining may have an impact on how much water a treatment plant can take in and how much it will cost to treat the water for human use [25]. People who obtain their water from downstream are mostly impacted by unfavorable activities at the upstream even if they are not directly involved in such activities [26]. Sand mining in rivers poses a threat to water supply hence the need to examine its impact.

This study's goal is to determine how sand mining affects the physicochemical, and biological characteristics of Volta Lake at Asutwae village and to assess the suitability of biochar produced from coconut husk in the removal of pollutants from water.

Methodology

Study area

The study was carried out in Asutwae village located in the Shai-Osudoku district in the Greater Accra Region of Ghana. It is in the southeast part of the country, between 5045' and 6005' degrees north latitude and 0005' east and 0020' west longitude (Fig. 1). The district covers up to 968 km² of the Greater Accra Region, or 29.84 %, in terms of land area (which covers a total of 3245 km²). Businesses, especially those looking for export prospects, benefit from its proximity to the Tema Metropolis and Accra [27].

Selection of the study area

A reconnaissance visit was made to identify potential sand mining sites on the Volta River. Three (3) sampling sites were randomly selected; upstream, midstream, and downstream. Upstream (N 06°05'41.4"; E 000°10'40.4") is located 425 m away from the sand mining site. This site was chosen as the reference site where there is no sand mining activity. Midstream (N 06°05'38.5"; E

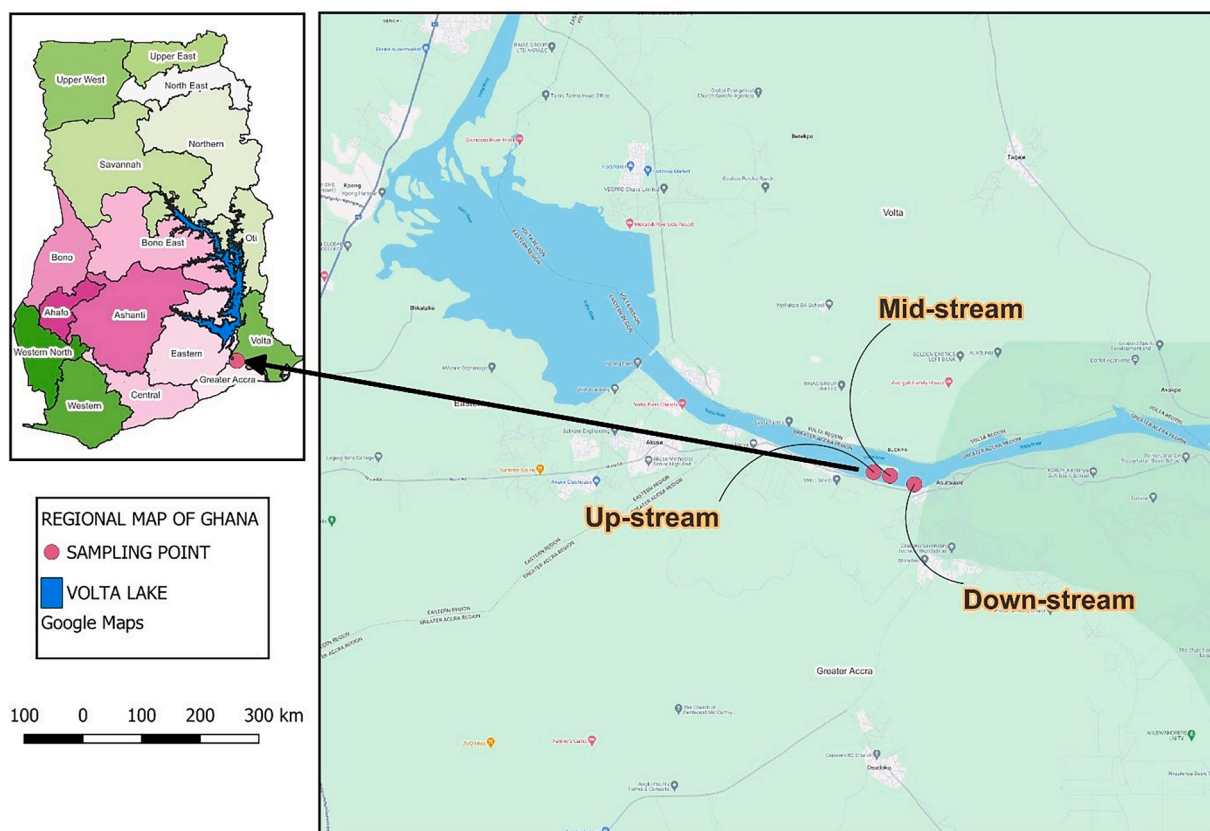


Fig. 1. Map showing the various sampling sites on Volta Lake, Ghana.

000°10'53.9") is the sand mining site, and downstream (N 06°05'31.3"; E 000°11'14.0") is located 656 m away from the sand mining site where a community water abstraction point is located. Water samples were collected at these sites for analysis, providing insights into how sand harvesting affects water quality.

Sampling procedure

Water sampling

On Volta Lake, twelve (12) water samples were taken at three (3) different locations. Four (4) samples each were collected from upstream (as reference), midstream (sand mining site), and downstream (close to water abstraction point) into a 1 L plastic bottle. A sampling point was established at a depth of 1 meter beneath the surface using a Van Dorn water sampler. Using a phosphate-free detergent, each sample bottle was thoroughly cleaned before being washed with a 1:1 solution of HNO₃, followed by distilled water, and then dried. Each plastic bottle was rinsed two times with the same water before collection. At each sampling point, Global Positioning System (GPS) coordinates were taken to map the locations. The samples were delivered to the laboratory for analysis after being kept in an ice chest. In situ measurements such as electrical conductivity, temperature, total dissolved solids (TDS), pH, and dissolved oxygen (DO) were done with the help of a multiparameter. Parameters such as Turbidity, Colour, Total Suspended Solids (TSS), Zinc (Zn), Lead (Pb), Manganese (Mn), Iron (Fe), Copper (Cu), Faecal Coliform (FC), Total Coliform (TC), *Escherichia Coli* (*E. coli*), Chemical Oxygen Demand (COD), and Biochemical Oxygen Demand (BOD) (Table 1), were analysed in the laboratory using standard laboratory methods [28].

Biochar production

The biochar for water treatment was made from coconut husk collected from roadside coconut sellers due to its abundance in the community. The biochar production procedure was adapted from Xiang et al. [29]. The collected coconut husk was washed with deionized water to remove sand and other unwanted debris after which it was air dried. The dried coconut husk was pyrolyzed at 850 °C and the coconut char was grind and sieved with 0.315 mm size mesh. The clean coconut husk obtained was dried at 120 °C in an oven to remove most contaminants and moisture. A carbonization process was used to decrease the volatile matter content of biochar using a flow rate of 0.65 litres per minute, a temperature of 850 °C, a time of 2 h, and a 10 °C/min heating rate. Elemental analyses were performed on the coconut husk to determine its properties. The activating agent was steam at a flow rate of 0.275 litres per minute. Two-thirds (14 cm) of the column (21 cm) were filled with media (biochar). An analysis of the midstream (sand mining site) water sample (1 L) was conducted after passing through the column (filtration).

Physicochemical analysis

Sample analysis

Statistical analysis

Data analysis was done using Inferential statistics, specifically the *t*-test, was employed to determine the statistical significance of the observed differences. At a 5 % confidence level ($p < 0.05$), the statistical test was verified as significant.

Treatment efficiency analysis

The overall removal efficiency was calculated as percentages using the equation:

$$\text{Overall removal efficiency (\%)} = ((C_x - C_y) / C_x) \times 100$$

Table 1
Analytical method employed.

Parameter	Methods
pH	pH Probe
Conductivity	Conductivity probe
Temperature	Temperature probe
Turbidity	Nephelometric Method
Colour (Apparent)	Visual comparison Method
Total Suspended Solids	Gravimetric Method
Metals (Fe, Mn, Cu, Zn)	Agilent Technologies 240FS AA
Dissolved Oxygen	YSI Pro DO meter
Chemical Oxygen Demand	Potassium dichromate Reflux Method (closed tube)
Biochemical Oxygen Demand	Dilution and Dissolved Oxygen determination after incubation at 20 °C for 5 days
Total Coliform	Membrane Filtration
Faecal Coliform	Membrane Filtration
<i>E. coli</i>	Membrane Filtration

where C_x is the initial concentration of water sample and C_y is the final concentration of water sample.

Results and discussions

Water quality

The results showed a variation in colour concentrations among water samples (Table 2). There was a statistically significant difference between water samples taken from, upstream and midstream (Table 3). The downstream, where a water abstraction point for the community is located also showed a statistically significant difference with midstream ($p = 0.00$). These findings suggest that sand mining alters the appearance of water by raising its colour concentration. Sand mining activities occurring midstream impact downstream where the water is being used for domestic purposes by the community. Particulates in the water from sand mining and other operations have increased the colour intensity. Previous research by Lekomo et al., [30] illustrated how sand mining affects the intensity of colour in river water lending support to the findings. Sand mining activities including stirring, digging, and scooping caused the water to become coloured and murky, which concord with Sreebha and Padmala., [31]; Madyise, [7]; and Yen & Rohasliney, [15].

The Turbidity value for upstream which is serving as a reference site ranged between 1.08 NTU and 1.38 NTU with a mean value of 1.20 NTU. The midstream (sand mining site) ranged between 8.69 NTU and 11.3 NTU with a mean of 10.1 NTU which was also above the WHO drinking water guideline. The downstream also recorded a range between 2.23 NTU and 3.23 NTU with a mean value of 2.78 NTU which was below the WHO drinking water guideline. An independent t -test analysis between the means of water samples upstream and midstream showed a significant difference of $p = 0.00$. Also, there were statistically significant differences between the water samples from midstream and downstream ($p = 0.00$) (Table 3).

This study results differed from those of Karikari et al., [32], which indicates that water clarity could deteriorate if left unmonitored. River sand mining increases suspended solids in the river, as a result, macroinvertebrate communities are affected by drift [33,34] stated that drifting invertebrates are important food sources for a variety of organisms, from fish to larger invertebrates. In addition to these findings, Yen and Rohasliney [15], and Ashraf et al., [16] found that sand mining operations generate water disturbances that led to high turbidity levels in those places. Sarkar, et al., [35] also found that turbid water can lead to gastrointestinal diseases in humans and impede the growth of submerged aquatic plants.

The mean electrical conductivity and TDS levels were acceptable according to WHO drinking water guidelines (Table 2), meaning dissolved ions were lower, thus lowering conductivity. Additionally, a separate sample t -test (Table 3) revealed that the mean difference between the water samples was not statistically significant ($p = 0.25$). However, there was a statistically significant difference in the mean between water samples upstream and midstream ($p = 0.02$). The low conductivity of the lake reflects its low ionic content Karikari et al., [32]. Research performed by Aris et.al., [36] has shown that high COD, high turbidity, and high BOD levels result in high TDS and conductivity in the water. Despite the notable differences in conductivity, a study by Daniel et al., [37] found that sand mining can increase dissolved ions and electrical conductivity, posing a health risk to humans and aquatic life.

The dissolved oxygen (DO) concentrations recorded were 5.05 mg/l, 4.99 mg/l, and 4.79 mg/l respectively for upstream, midstream, and downstream (Table 2). An independent sample t -test revealed the difference in mean between the water samples taken upstream and midstream was statistically insignificant. However, there was a statistically significant difference in the mean between water samples in midstream and downstream (Table 3). This implies that sand mining occurring at midstream can affect the DO concentration downstream. Bhattacharya, [38] indicated that mining intensity can significantly fluctuate DO concentration in a water body. The results of this study were different from Karikari et al., [32] where they recorded a mean DO concentration between 7.30 mg/l. They also stated that the Volta Lake supports aquatic life since its oxygen supply is affected by water turbulence and wind action.

Table 2
Water quality data for the sampling sites alongside the WHO drinking water guideline.

Parameter	Units	Upstream	Midstream	Downstream	WHO guideline value
Temperature	°C	28.3	28.4	28.5	–
pH	pH Units	7.19	7.04	7.07	6.50–8.50
Conductivity	µS/cm	72.3	75.2	73.7	–
Turbidity	NTU	1.20	10.1	2.78	5.00
Colour	Hazen	2.50	11.3	5.63	15.0
Total Suspended Solids	mg/l	1.00	8.50	2.50	–
Total Dissolved Solids	mg/l	43.4	45.1	44.2	1000
Dissolved Oxygen	mg/l	5.05	4.99	4.79	–
Biochemical Oxygen Demand	mg/l	1.78	8.30	7.35	–
Chemical Oxygen Dissolved	mg/l	9.97	43.1	31.6	–
Iron	mg/l	0.031	0.160	0.052	0.300
Manganese	mg/l	0.050	0.303	0.097	0.400
Zinc	mg/l	<0.005	<0.005	<0.005	2.00
Copper	mg/l	<0.010	<0.010	<0.010	2.00
Lead	mg/l	<0.005	<0.005	<0.005	0.01
Total Coliform	cfu/100ml	0	381	977	0
Faecal Coliform	cfu/100ml	0	0	0	0
E. coli	cfu/100ml	0	0	0	0

Table 3
Inferential statistics for the sampling sites.

Parameter	Units	Mean value			t-Test (p-value)		
		Upstream (US)	Midstream (MS)	Downstream (DS)	US-MS	MS-DS	US-DS
Colour	Hazen	2.50	11.3	5.63	0.00	0.01	0.00
Turbidity	NTU	1.20	10.1	2.78	0.00	0.00	0.00
pH	pH Units	7.19	7.04	7.07	0.06	0.45	0.08
Temperature	°C	28.3	28.4	28.5	0.39	0.39	0.13
Conductivity	µS/cm	72.3	75.2	73.7	0.02	0.25	0.29
Total Suspended Solids	mg/l	1.00	8.50	2.50	0.00	0.00	0.00
Total Dissolved Solids	mg/l	43.4	45.1	44.2	0.02	0.25	0.29
Dissolved Oxygen	mg/l	5.05	4.99	4.79	0.55	0.02	0.01
Biochemical Oxygen Demand	mg/l	1.78	8.30	7.35	0.00	0.00	0.00
Chemical Oxygen Dissolved	mg/l	9.97	43.1	31.6	0.00	0.00	0.00
Iron	mg/l	0.031	0.160	0.052	0.00	0.00	0.00
Manganese	mg/l	0.050	0.303	0.097	0.00	0.00	0.00
Zinc	mg/l	0.000	0.001	0.000	0.36	0.36	0.36
Copper	mg/l	0.000	0.000	0.000	–	–	–
Lead	mg/l	0.000	0.000	0.000	–	–	–

The BOD levels at midstream and downstream were all higher than those recorded upstream (Table 2). An independent sample *t*-test was done and recorded a statistically significant difference in the means of the water samples (Table 3). However, it is evident from the results of the analysis that the midstream and downstream had higher BOD levels than the upstream. Consequently, the abundance of biodegradable material in the water suggests that it has been polluted. Tay [39] stated that depending on the source, contamination may have come from human activities both outside and inside the river. According to Bhateria and Jain [40], increased amounts of BOD in the water were shown to be caused by increased organic waste from leaves, dead plants, dump sites and runoff from surrounding farms. The aquatic ecology is put in peril by low oxygen levels because they reduce productivity. Aris et al., [36] found high amounts of BOD result from sand mining within rivers and adjacent areas, making them unusable for humans and dangerous for other organisms reliant on the river

The mean COD values for upstream (reference site) were lower than both midstream (sand mining) and downstream (no sand mining). The COD scores after an independent sample *t*-test showed a statistically significant difference in the means of water samples from the mining sites and the non-sand mining sites (Table 3). These findings point to a rise in COD concentrations at the midstream and downstream, which may be related to sand mining activities in that river section. It might also be driven by human activities both inside and outside the river that create pollution and the breakdown of organic substances downstream. An earlier study by Kolawole et al., [41] showed the same results with COD levels of 58.40 mg/l in Asa River, Nigeria, brought about by both inorganic and organic pollutants. Aris et al., [36] also found high levels of COD in selected Malaysian rivers linked to sand mining.

The heavy metals concentrations of the water samples were below the WHO guideline values. The mean iron and manganese concentrations were a bit higher at midstream than upstream and downstream. Independent samples *t*-tests showed statistically significant variations in the means of water samples from the mining site (midstream) and the non-sand mining sites (upstream and downstream). Daniel et al., [37] stated that sand mining sites have higher levels of heavy metals compared to non-sand mining sites, indicating that sand mining was a factor in the high heavy metals level. In River Likangala, Malawi, Ullberg [42] also found iron and manganese concentrations exceeding WHO drinking water guideline. It was also noted that copper, zinc, and lead levels for upstream, midstream, and downstream were below the WHO drinking water guideline. It is interesting to note that this result differs with that of Ullberg [42], whose study discovered trace quantities of copper, zinc, and lead in the river water despite anthropogenic activity.

Total coliforms, faecal coliforms, and *E. coli* were not found in the upstream area (reference site). The midstream recorded a mean

Table 4
Major and minor oxides of coconut husk biochar.

Major Oxides		Minor Oxides	
Sample ID	Coconut Husk	Sample ID	Coconut Husk
%MgO	2.35	V ₂ O ₅ /ppm	0
%Al ₂ O ₃	0.887	Cr ₂ O ₃ /ppm	1004
%SiO ₂	2.38	Rb ₂ O/ppm	125
%P ₂ O ₅	0.649	SrO/ppm	44.8
%SO ₄	0.667	Y ₂ O ₃ /ppm	0
%SO ₃	0.556	ZrO ₂ /ppm	35.9
%K ₂ O ₅	10.2	Nb ₂ O ₅ /ppm	7.72
%K ₂ O	6.06	SnO ₂ /ppm	28.8
%CaO	0.540	WO ₃ /ppm	0
%TiO ₂	0	ThO ₂ /ppm	16.8
%MnO	0.016	U ₃ O ₈ /ppm	0
%Fe ₂ O ₃	1.26		

total coliform count but no faecal coliform or *E. coli*. The downstream also recorded a mean total coliform which was higher compared to the midstream (Table 2). The high level of total coliform in the downstream may be due to human settlement due to sand mining. As a result of these findings, the microbial population of the lake may be impacted by sand mining activities.

Biochar characterization

To determine the composition of the ashes, X-ray fluorescence spectroscopy (XRF) was used. An XRF analysis determines the percentages of elements in ashes. A variety of elements, including MgO, Fe₂O₃, SiO₂, CaO, K₂O, and K₂O₅, coexist in biochar (Table 4).

Overall removal efficiency

Coconut husk biochar was examined for its ability to remove pollutants from water. A wide range of pollutants was successfully treated with coconut husk biochar. A higher removal efficiency was achieved with turbidity (91.0 %), TSS (94.2 %), Total coliform (98.6 %), and Manganese (99.6 %), compared to 87.5 %, 86.9 %, 70.8 %, and 73.3 % for colour, iron, BOD, and COD, respectively (Table 5).

While the selected treatment methods made a significant difference in this study, they may be more effective in removing other pollutants, such as dyes or metals [43].

Possible removal mechanisms

The use of biochar as an adsorbent has shown exceptionally high removal rates (up to 100 %), depending on the status of the contaminants. Li et al., [44] reported that biochar has an abundance of negatively and positively charged sites that can bind cations and anions to them. The interaction between biochar surface and ions in the solution may be influenced by both physical and chemical interactions, including electrostatic interaction, dispersive forces, precipitation, and complexation [21]. Beiyan et al., [45] found that biochar was more effective for heavy metal removal than limestone and apatite as a result of the deprotonation of acidic functional groups on biochar surfaces at high pH. Karunanayake et al. [46] also reported these cations from the biochar attract positively charged metals such as Cd²⁺ and Pb²⁺. [47] proposed that bacterial adhesion on biochar surfaces might be linked to attractive van der Waals interactions and repulsive electrostatic interactions following the Derjaguin-Landau-Verwey-Overbeek theory.

Conclusion and recommendation

Conclusion

The excavation of in-stream sand has detrimental environmental impacts, such as fading water clarity. A significant contribution to this change can be attributed to the sand mining activities carried out in the river. Several water quality parameters of the river, including its colour, turbidity, TSS, BOD, and COD levels, as well as its iron and manganese concentrations, were altered negatively by sand mining activities. Water with lower quality also presents a threat to aquatic life, therefore, it is not just an issue for human health. Although sand mining on the Volta River has socioeconomic benefits, such as creating jobs and generating income, indiscriminate sand mining negatively impacts the environment and water quality.

The study showed that coconut husk biochar is an effective sorbent for removing pollutants from water samples (>70 % removal efficiency).

Recommendation

Considering the increasing practice of in-stream sand mining, sand miners and residents should be urged to utilize the bar skimming technique because it has little influence on the water quality. Because sand mining frequently results in open dangerous pits that pose threats to the environment and human health, it is crucial to restore these regions.

The physico-chemical and microbial parameters of the water where extraction is ongoing should be regularly monitored to assess their utility and aquatic life compatibility.

A comprehensive awareness and public education campaign for all stakeholders is required to ensure support and effective implementation of sand regulating policies.

The management of waste always poses economic, socio-political, and environmental challenges, which make low-cost technology for water treatment even more appealing.

It would be useful to conduct further research to determine the impacts of sand mining on the macroinvertebrate population on Volta Lake.

CRedit authorship contribution statement

Daniel Kwaku Amoah: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Resources, Writing – original draft. **Esi Awuah:** Project administration, Supervision, Writing – review & editing. **Isaac O.A. Hodgson:** Supervision, Writing – review & editing. **Miriam Appiah-Brempong:** Supervision, Writing – review & editing. **Elizabeth Von-Kiti:** Investigation, Resources, Writing – review & editing. **Isaac Nii Yartey Nyarko:** Resources.

Table 5

Removal efficiency of water pollutants using coconut husk biochar.

Parameter	Initial concentrations of water sample (Cx)	Final concentrations of water sample (Cy)	Removal efficiency (%)
Turbidity	18.1	1.62	91.0
Colour	20.0	2.50	87.5
Total Suspended Solids	17.2	1.00	94.2
Iron	0.337	0.044	86.9
Manganese	0.522	0.002	99.6
Biochemical Oxygen Demand	9.60	2.80	70.8
Chemical Oxygen Dissolved	48.0	12.8	73.3
Total Coliform	558	8.00	98.6

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] [48] WHO and UNICEF, World Health Organization. Progress on Drinking water, sanitation, and hygiene: 2017 Update and SDG Baselines, World Health Organization (WHO) and the United Nations Children's Fund (UNICEF), Geneva, Switzerland, 2017.
- [2] S.-Y. Lu, H.-M. Zhang, S.O. Sojinu, G.-H. Liu, J.-Q. Zhang, H.-G. Ni, Trace elements contamination and human health risk assessment in drinking water from Shenzhen, China, *Environ. Monitor. Assess.* 187 (1) (2015) 4220.
- [3] D. Chakraborti, M.M. Rahman, S. Ahamed, R.N. Dutta, S. Pati, S.C. Mukherjee, Arsenic groundwater contamination and its health effects in Patna district (capital of Bihar) in the middle Ganga plain, India. *Chemosphere* 152 (2016) 520–529.
- [4] M.N. Saviour, Environmental impact of soil and sand mining: a review, *Int J Sci Environ Technol* 1 (3) (2012) 125–134.
- [5] M. Mandal, D. Ghosh, B. Ghosh, Environmental impact of sand mining: a case study along the lower reaches of Ajay River, West Bengal, India, *J. Environ. Sociobiol.* (2016) 99–108.
- [6] Arkoh Isaac, (2019). Sand winning threatens coastal tourism in Central Region. Retrieved from <https://www.modernghana.com/news/974948/sand-winning-threatens-coastal-tourism-in-central.html>.
- [7] T. Madyise, Case Studies of Environmental Impacts of Sand Mining and Gravel Extraction For Urban Development in Gaborone (Doctoral dissertation, University of South Africa, 2013).
- [8] D. Padmalal, K. Maya, Sand mining: Environmental Impacts and Selected Case Studies, Springer, 2014.
- [9] L. Koehnken, M. Rintoul, Impacts of sand mining on ecosystem structure, process and biodiversity in rivers, *World Wildlife Fund International* (2018) 159.
- [10] S. Damseth, K. Thakur, R. Kumar, S. Kumar, D. Mahajan, H. Kumari, A.K. Sharma, Assessing the Impacts of River Bed Mining On Aquatic ecosystems: A critical Review of Effects On Water Quality and Biodiversity, *HydroResearch*, 2024.
- [11] M.R. Mahananda, B.P. Mohanty, N.R. Behera, Physico-chemical analysis of surface and groundwater of Bargarh District Orissa, India, *Int. J. Res. Rev. Appl. Sci.* 2 (3) (2010) 284–295.
- [12] K. Venkatesharaju, R.K. Somashekar, K.L. Prakash, Study of seasonal and spatial variation in surface water quality of Cauvery River stretch in Karnataka, *J. Ecol. Nat. Environ.* 2 (1) (2010) 001–009.
- [13] H.M. Pereira, L.M. Navarro, I.S. Martins, Global biodiversity change: the bad, the good, and the unknown, *Annu Rev. Environ. Resour.* 37 (2012) 25–50.
- [14] R.K. Bhattacharya, N. Das Chatterjee, G. Dolui, Consequences of sand mining on water quality and instream biota in alluvial stream: a case-specific study in South Bengal River, India, *Sustain. Water Resour. Manag.* 5 (2019) 1815–1832.
- [15] T.P. Yen, H. Rohasliney, Status of water quality subject to sand mining in the Kelantan River, Kelantan, *Trop. Life Sci. Res.* 24 (1) (2013) 19.
- [16] M.A. Ashraf, M.J. Maah, I. Yusoff, A. Wajid, K. Mahmood, Sand mining effects, causes and concerns: a case study from Bestari Jaya, Selangor, Peninsular Malaysia, *Sci. Res. Essays* 6 (6) (2011) 1216–1231.
- [17] Z. Gokalp, S. Bulut, Potential use of biochar in wastewater treatment operations and soil improvement, *Curr. Trends Nat. Sci.* 11 (21) (2022) 161–169.
- [18] M. Guo, The 3R principles for applying biochar to improve soil health, *Soil. Syst.* 4 (1) (2020) 9.
- [19] P. Rao, V. Rathod, Valorization of food and agricultural waste: a step towards greener future, *Chem. Rec.* 19 (9) (2019) 1858–1871.
- [20] Y. Wang, H. Li, S. Lin, Advances in the study of heavy metal adsorption from water and soil by modified biochar, *Water. (Basel)* 14 (23) (2022) 3894.
- [21] H. Jin, M.U. Hanif, S. Capareda, Z. Chang, H. Huang, Y. Ai, Copper (II) removal potential from aqueous solution by pyrolysis biochar derived from anaerobically digested algae-dairy-manure and effect of KOH activation, *J. Environ. Chem. Eng.* 4 (1) (2016) 365–372.
- [22] K.N. Palansooriya, Y. Yang, Y.F. Tsang, B. Sarkar, D. Hou, X. Cao, Y.S. Ok, Occurrence of contaminants in drinking water sources and the potential of biochar for water quality improvement: a review, *Crit. Rev. Environ. Sci. Technol.* 50 (6) (2020) 549–611.
- [23] W. Gwenzi, N. Chaukura, C. Noubactep, F.N. Mukome, Biochar-based water treatment systems as a potential low-cost and sustainable technology for clean water provision, *J. Environ. Manage.* 197 (2017) 732–749.
- [24] E. Weidner, E. Karbassiyazdi, A. Altaee, T. Jesionowski, F. Ciesielczyk, Hybrid metal oxide/biochar materials for wastewater treatment technology: a review, *ACS. Omega* 7 (31) (2022) 27062–27078.
- [25] L.I. Nasare, S.A. Opoku, A. Amponsah, D. Tom-Dery, W.J. Asante, B.N. Baatuwile, Effect of sand mining on riparian landcover transformation in Dallung-Kukou catchment of the White Volta basin, Ghana. *Heliyon* 9 (8) (2023).
- [26] E. Bessah, A.O. Raji, O.J. Taiwo, S.K. Agodzo, O.O. Ololade, A. Strapasson, E. Donkor, Assessment of surface waters and pollution impacts in Southern Ghana, *Hydro. Res.* 52 (6) (2021) 1423–1435.
- [27] Ghana Statistical Service (GSS). 2010 Population and housing census, housing in Ghana, 2014. <http://www.statsghana.gov.gh/gssmain/fileUpload/pressrelease/Housing%20in%20Ghana.pdf>.
- [28] R.B. Baird, A.D. Eaton, E.W. Rice, L. Bridgewater, Standard Methods For the Examination of Water and Wastewater, American Public Health Association, 2017.
- [29] W. Xiang, X. Zhang, J. Chen, W. Zou, F. He, X. Hu, D.C.W. Tsang, Y.S. Ok, B. Gao, Biochar technology in wastewater treatment: a critical review, *Chemosphere* 252 (2020) 126539.
- [30] Y.K. Lekomo, C.M. Ekengoue, A. Doula, R.F. Lele, G.C. Suh, S. Obiri, A.K. Dongmo, Assessing impacts of sand mining on water quality in Toutsang locality and design of wastewater purification system, *Clean.Eng. Technol.* 2 (2021) 100045, <https://doi.org/10.1016/j.clet.2021.100045>.
- [31] S. Sreebha, D. Padmalal, Environmental impact assessment of sand mining from the small catchment rivers in the southwestern coast of India: a case study, *Environ. Manage* 47 (2011) 130–140.
- [32] A.Y. Karikari, F. Akpabey, E.K. Abban, Assessment of water quality and primary productivity characteristics of Volta Lake in Ghana. *Acad. J. Environ. Sci.* 1 (5) (2013) 088–103.

- [33] G.G. Gitunja, Impacts of Land Use and Land Cover On Water Quality and Benthic Macroinvertebrates in Theta River Catchment (Doctoral Dissertation, University of Nairobi, 2020).
- [34] L. Koehnken, M.S. Rintoul, M. Goichot, D. Tickner, A.C. Loftus, M.C. Acreman, Impacts of riverine sand mining on freshwater ecosystems: a review of the scientific evidence and guidance for future research, *River. Res. Appl.* 36 (3) (2020) 362–370.
- [35] A.M. Sarkar, A.K.M.L. Rahman, A. Samad, A.C. Bhowmick, J.B. Islam, Surface and groundwater pollution in Bangladesh: a review, *Asian Rev. Environ. Earth Sci.* 6 (1) (2019) 47–69.
- [36] A.Z. Aris, W.Y. Lim, S.M. Praveena, M.K. Yusoff, M.F. Ramli, H. Juahir, Water quality status of selected rivers in Kota Marudu, Sabah, Malaysia, and its suitability for usage, *Sains. Malays* 43 (3) (2014) 377–388.
- [37] I.K. Daniel, G.N. Kamau, J.K. Mbugua, Assessment of the effects of sand harvesting on river water quality: case study in Iuuma river, kivaa ward, machakos county, MOJ. *Proteom. Bioinform.* 6 (2017) 195–200.
- [38] R.K. Bhattacharya, Instream sand mining impact on water quality and benthos community in an alluvial reach: a case study on river Kangsabati, West Bengal, *International Journal of Current Research in Life Sciences* 7 (8) (2018) 2613–12617.
- [39] C.K. Tay, Integrating water quality indices and multivariate statistical techniques for water pollution assessment of the Volta Lake, Ghana, *Sustain. Water Resour. Manag.* 7 (5) (2021) 71.
- [40] R. Bhateria, D. Jain, Water quality assessment of lake water: a review, *Sustain. Water. Resour. Manage* 2 (2016) 161–173.
- [41] O.M. Kolawole, K.T. Ajayi, A.B. Olayemi, A.I. Okoh, Assessment of water quality in Asa River (Nigeria) and its indigenous *Clarias gariepinus* fish, *Int. J. Environ. Res. Public Health* 8 (11) (2011) 4332–4352.
- [42] M. Ullberg, Temporal Water Quality Study of the Heavily Human-Impacted Likangala River, Zomba, Malawi, 2015.
- [43] T. Halfhide, L.J. Lalgee, K.S. Singh, J. Williams, M. Sealy, A. Manoo, A. Mohammed, Nutrient removal using spent coconut husks, *h2oj* 2 (1) (2019) 125–136.
- [44] X. Li, C. Zhao, M. Zhang, Chapter 8 - Biochar for anionic contaminants removal from water, in: Y.S. Ok, D.C.W. Tsang, N. Bolan, J.M. Novak (Eds.), *Biochar from Biomass and Waste*, Elsevier, Amsterdam, Netherlands, 2019, pp. 143–160.
- [45] J. Beiyuan, D.C. Tsang, A.C. Yip, W. Zhang, Y.S. Ok, X.-D. Li, Risk mitigation by waste-based permeable reactive barriers for groundwater pollution control at e-waste recycling sites, *Environ. Geochem. Health* 39 (1) (2017) 75–88, <https://doi.org/10.1007/s10653-016-9808-2>.
- [46] A.G. Karunanayake, O.A. Todd, M.L. Crowley, L.B. Ricchetti, C.U. Pittman, R. Anderson, T.E. Mlsna, Rapid removal of salicylic acid, 4-nitroaniline, benzoic acid, and phthalic acid from wastewater using magnetized fast pyrolysis biochar from waste Douglas fir, *Chem. Eng. J.* 319 (2017) 75–88, <https://doi.org/10.1016/j.cej.2017.02.116>.
- [47] A.Y. Lau, D.C. Tsang, N.J. Graham, Y.S. Ok, X. Yang, X.-D. Li, Surface modified biochar in a bioretention system for *Escherichia coli* removal from stormwater, *Chemosphere* 169 (2017) 89–98, <https://doi.org/10.1016/j.chemosphere.2016.11.048>.